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Dissecting the Effect of Credit Supply on Trade: Evidence from Matched Credit–Export Data

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1 What the paper is doing

1.1 Question

The paper asks a causal question:

How does a negative shock to *bank credit supply* affect firms' exports, and through which margins (intensive vs. extensive)?

1.2 Core difficulty

Credit C_{it} is an *equilibrium object*. A fall in credit could reflect:

- a fall in *credit supply* (banks' ability/willingness to lend), or
- a fall in *credit demand* (firms cut desired borrowing when exports fall for non-credit reasons).

Therefore, regressions of exports on observed credit changes generally confound demand and supply.

1.3 Core idea

The paper uses the 2008 financial crisis as a quasi-experimental episode in *Peru*: foreign bank funding to the Peruvian banking system drops sharply, and banks that relied more on foreign funding experience a larger contraction in lending. Using matched bank–firm credit registry data and customs export data, the authors build a firm-level instrument for credit supply shocks based on:

1. banks' *pre-crisis* foreign-funding dependence, and
2. firms' *pre-crisis* exposure to those banks through lending relationships.

1.4 Main results (economic interpretation)

- **Intensive margin:** exports of continuing firm–product–destination flows respond positively to credit; the IV elasticity is economically meaningful.
- **Extensive margin:** entry/exit of firm–product–destination export relationships is largely unaffected.
- **Mechanism:** credit mainly affects *variable/export operating decisions* (shipment frequency/size), not the *sunk* decisions associated with entering new export markets.
- **Methodological warning:** using aggregated outcomes (firm- or sector-level exports) without controlling for product–destination shocks can produce severely biased estimates.

2 Data structure and notation

2.1 Indexing and time aggregation

- i : firm (exporter).
- b : bank.
- p : product (baseline: 4-digit HS).
- d : destination country.
- t : time. The analysis collapses monthly data into two annual windows:

$$t \in \{\text{Pre}, \text{Post}\},$$

where **Pre** is the 12 months before July 2008 and **Post** is the 12 months after July 2008.

2.2 Key variables

- Exports at the flow level:

$$X_{ipdt} \equiv \text{exports by firm } i \text{ of product } p \text{ to destination } d \text{ in period } t.$$

The baseline outcome uses *export volume* (kg), which reduces concerns about price variation.

- Firm credit:

$$C_{it} \equiv \text{total outstanding bank credit to firm } i \text{ in period } t \text{ (sum across banks).}$$

- Firm–bank credit:

$$C_{ibt} \equiv \text{outstanding credit of firm } i \text{ with bank } b \text{ in period } t.$$

- Bank foreign-funding dependence (pre-crisis):

$$FD_b \equiv \frac{\text{foreign liabilities of bank } b}{\text{assets of bank } b},$$

measured at year-end 2006 (predetermined).

2.3 Margins of trade

The paper defines margins *at the firm–product–destination level*:

- **Intensive margin**: changes in X_{ipdt} *conditional on* the flow being active in both Pre and Post.
- **Extensive margin**: changes in the *indicator* of whether the flow exists (entry/exit).

This definition matters because it allows control for product–destination shocks (international demand, input prices, transport costs) at a granular level.

3 Conceptual model and why OLS is biased

3.1 Export “reduced form”

The paper starts from a generic representation:

$$X_{ipdt} = X_{ipdt}(H_{ipdt}, C_{it}), \quad (3.1)$$

where H_{ipdt} collects non-credit determinants of exports (demand in d for p , production costs, productivity, etc.).

3.2 Credit as an equilibrium object

Credit itself depends on demand and supply:

$$C_{it} = C_{it}(H_{ipdt}, \dots, S_{it}), \quad (3.2)$$

where S_{it} is the credit supply faced by firm i . Because H_{ipdt} affects both exports and credit, OLS of exports on credit is generally biased.

3.3 Elasticity object of interest

The target is the elasticity of exports with respect to credit (intensive margin):

$$\eta \equiv \frac{\partial X_{ipdt}}{\partial C_{it}} \cdot \frac{C_{it}}{X_{ipdt}}.$$

Empirically, the paper estimates log specifications.

4 Baseline empirical models (levels)

4.1 Intensive margin: flow-level log specification

The intensive margin elasticity is estimated from:

$$\ln(X_{ipdt}) = \eta^I \ln(C_{it}) + \delta_{ipd} + \alpha_{pdt} + \varepsilon_{ipdt}, \quad (3.3)$$

where:

- δ_{ipd} is a firm–product–destination fixed effect (time-invariant heterogeneity in the match),
- α_{pdt} is a product–destination–time effect capturing demand and cost shocks common to all firms exporting p to d at t ,
- ε_{ipdt} is an idiosyncratic residual.

Why this structure is central. By conditioning on (p, d, t) , identification compares firms exporting the *same product to the same destination* at the same time, which substantially reduces confounding from non-credit shocks.

4.2 Extensive margin: linear probability model

Let E_{ipdt} be an indicator for exporting product p to destination d at time t . The extensive margin specification is:

$$E_{ipdt} = \eta^E \ln(C_{it}) + \delta_i + \alpha_{pdt} + \varepsilon_{ipdt}. \quad (3.4) \quad (4)$$

Entry vs. exit measurement. The paper estimates (3.4) separately for:

- **Exit:** $E_{ipdt} = 1$ if the flow is *inactive* in t conditional on being active in $t - 1$.
- **Entry:** $E_{ipdt} = 1$ if the flow is *active* in t conditional on being inactive in $t - 1$.

Fixed effects choice. Unlike the intensive margin, the extensive margin uses firm fixed effects δ_i rather than firm–product–destination fixed effects, because the same (i, p, d) observation cannot both enter and exit in adjacent periods. The resulting coefficient η^E is best interpreted as a *semi-elasticity* of entry/exit probabilities with respect to $\ln C_{it}$.

5 Identifying the credit supply shock

The paper’s identification strategy has two connected pillars:

1. show that banks more exposed to foreign funding suffered a larger credit contraction after July 2008, and
2. use firms’ exposure to such banks (predetermined) to instrument firm credit changes.

5.1 Within-firm bank lending regression

Define C_{ibt} as firm–bank credit in the two periods $t \in \{\text{Pre}, \text{Post}\}$. The paper estimates:

$$\ln(C_{ibt}) = \theta_{ib} + \mu_{it} + \beta f(FD_b) \times \text{Post}_t + \nu_{ibt}, \quad (3.5) \quad (5)$$

where:

- θ_{ib} (firm–bank FE) absorbs time-invariant matching between firm i and bank b ,
- μ_{it} (firm–time FE) absorbs changes in firm-level credit demand between Pre and Post,
- $f(FD_b)$ is a (possibly nonlinear) function of bank foreign-funding dependence,
- Post_t is an indicator for the Post period.

Identification intuition. Within the same firm i , μ_{it} absorbs the firm’s overall change in desired borrowing. Thus β is identified from *differential changes across the firm’s lenders* as a function of FD_b .

5.2 First-differenced bank-lending equation

First-differencing (3.5) eliminates θ_{ib} :

$$\ln(C_{ib,Post}) - \ln(C_{ib,Pre}) = \mu'_i + \beta f(FD_b) + \nu_{ib}, \quad (4.8) \quad (6)$$

where $\mu'_i \equiv \mu_{i,Post} - \mu_{i,Pre}$ is a firm dummy absorbing changes in firm credit demand.

5.3 Instrument construction: shift-share exposure

The firm-level exposure to the bank shock is:

$$F_i = \sum_b \omega_{ib} f(FD_b), \quad (3.6) \quad (7)$$

where ω_{ib} is bank b 's share in firm i 's total credit (measured pre-crisis, year-end 2006).

To generate time variation, interact with $Post_t$:

$$F_{it} = F_i \cdot Post_t. \quad (3.7) \quad (8)$$

Why predetermined components matter. Both FD_b and ω_{ib} are measured before the crisis, reducing concerns that bank exposure is chosen in anticipation of the shock.

5.4 Choosing the functional form $f(\cdot)$ via LASSO

The paper considers alternative parametrizations of $f(FD_b)$ (binary indicator, linear, polynomial). To avoid ad hoc choice, it uses LASSO-type selection (Belloni et al., 2012) to pick a functional form that best predicts lending contraction.

The baseline instrument is a *binary high-exposure indicator*:

$$F_i = \sum_b \omega_{ib} D(FD_b > 10\%), \quad (4.10) \quad (9)$$

i.e., the share of firm credit coming from banks whose foreign funding share exceeds 10% (the mean among commercial banks).

6 2SLS implementation in the collapsed two-period data

Once the data are collapsed to Pre/Post, the intensive margin is estimated in first differences.

6.1 First stage for firm credit

The first stage is:

$$\ln(C_{i,Post}) - \ln(C_{i,Pre}) = \gamma'_{pd} + \alpha F_i + \varepsilon_i, \quad (4.9) \quad (10)$$

where γ'_{pd} controls for product–destination composition through fixed effects and F_i is defined in (4.10).

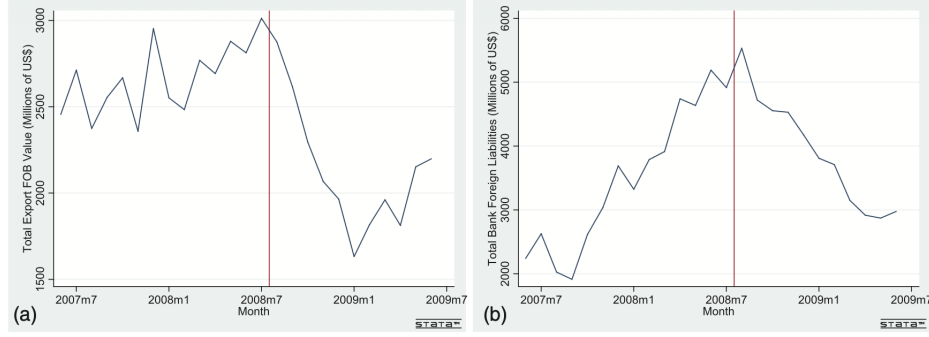


FIGURE 1

6.2 Second stage for the intensive margin

Differencing (3.3) eliminates δ_{ipd} and yields:

$$\ln(X_{ipd,Post}) - \ln(X_{ipd,Pre}) = \alpha'_{pd} + \eta^I [\ln(C_{i,Post}) - \ln(C_{i,Pre})] + \varepsilon'_{ipd}. \quad (4.11) \quad (11)$$

The key regressor is firm-level credit growth; the key control is the product–destination shock α'_{pd} ; and the key instrument is F_i .

6.3 Extensive margin estimation

Entry/exit are not differenced (because they are changes in export status), so the extensive margin is estimated in levels using (3.4), instrumenting $\ln C_{it}$ with $F_i \times Post_t$.

7 Figures: what they show and how they connect to identification

7.1 Figure 1: aggregate crisis timing

What is plotted. Total Peruvian exports and total banking-sector foreign financing (foreign liabilities) over time, with a vertical line at July 2008.

Interpretation. The crisis is visible in both macro series: foreign financing collapses and exports decline. The empirical design uses July 2008 as the break defining Pre vs. Post.

7.2 Figure 2: bank credit contraction vs. foreign-funding dependence

Panel (a). The change in total bank credit (log change between Post and Pre) is plotted against FD_b .

Panel (b). The change in *credit supply* is plotted after removing firm demand components (residual from a regression with firm FE), again against FD_b .

Interpretation. Both panels show that higher FD_b predicts larger lending contractions, supporting the relevance of the bank foreign-funding channel.

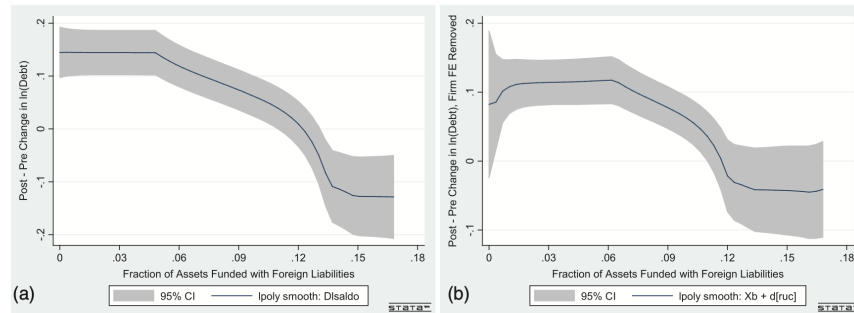


FIGURE 2
Bank credit and foreign funding. (a) Change in total credit; (b) Change in credit supply.

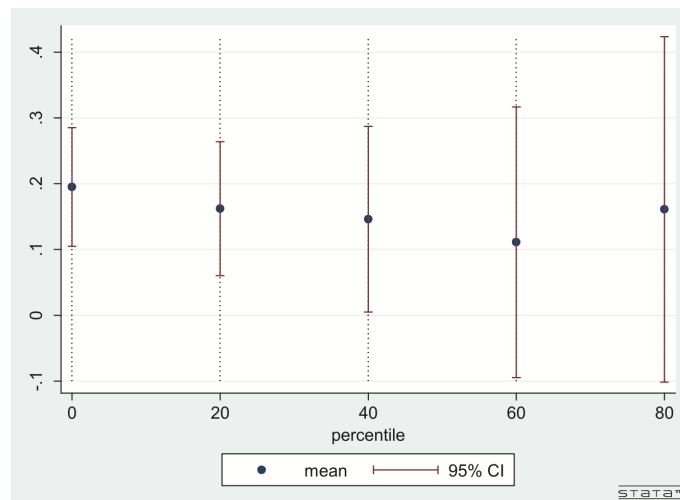


FIGURE 3
Heterogeneity of the export elasticity to credit.

TABLE 1
Bank descriptive statistics

	All financial institutions			Commercial banks						S&L institutions		
	(N=41)			High foreign exposure (N=4)			Low foreign exposure (N=9)			(N=28)		
	Mean (1)	SD (2)	p50 (3)	Mean (4)	SD (5)	p50 (6)	Mean (7)	SD (8)	p50 (9)	Mean (10)	SD (11)	p50 (12)
Assets (M US\$)	1102	2744	134	2533	3817	794	3559	4568	1023	108	108	62
Loans/assets	0.698	0.178	0.756	0.659	0.126	0.660	0.582	0.189	0.601	0.741	0.167	0.773
Deposits/assets	0.586	0.197	0.642	0.573	0.082	0.543	0.696	0.147	0.735	0.553	0.212	0.621
Foreign financing/assets	0.055	0.108	0.002	0.196	0.135	0.175	0.050	0.034	0.065	0.037	0.108	0.000

7.3 Figure 3: heterogeneity of intensive-margin elasticities

What is plotted. The IV estimate of η^I for subsamples of export flows with different predicted continuation probabilities (cutoffs at different percentiles).

Interpretation. Point estimates are similar across subsamples; confidence intervals widen for stricter cutoffs (smaller samples). This supports the claim that the intensive-margin LATE is representative of the broader set of export flows.

8 Tables: models tested and how to read the results (Table-by-table)

8.1 Table 1: bank descriptive statistics

Purpose. Document heterogeneity in banks' foreign financing before the crisis and define "high" vs. "low" foreign exposure.

Reading.

- Banks with high foreign exposure are larger (assets) and rely more on foreign liabilities.
- The threshold for "high exposure" is based on whether FD_b is above the mean among commercial banks (around 10%).
- This cross-bank heterogeneity is the raw material for identification.

8.2 Table 2: firm and export-flow descriptive statistics

Purpose. Describe exporters and the firm-product-destination flow unit.

Panel 1 (firm-level).

- Shows typical exporter debt levels, number of lenders, fraction of dollar debt, and export scale.
- Highlights that many firms have multiple banking relationships; this matters because the within-firm estimator (3.5) relies on variation across a firm's lenders.

Panel 2 (flow-level).

TABLE 2
Firm descriptive statistics

	All exporters		Full subsample		Difference of means (3)–(1)		Subsample More than 1 bank	
	Mean (1)	SD (2)	Mean (3)	SD (4)	Diff (5)	t-test (6)	Mean (7)	SD (8)
Panel 1: Firm	(N = 6169)		(N = 4974)				(N = 2223)	
Debt (1000 US\$)	1013	6903	1253	7668	239	0.087	2753	11,286
No. of Lenders	1.70	1.10	1.75	1.12	0.05	0.032	2.29	1.21
Fraction of Dollar Debt	0.708	0.385	0.713	0.381	0.005	0.631	0.751	0.345
Exports - FOB (1000 US\$)	3348	52,721	4005	58,478	657	0.538	7510	82,027
Exports (1000 kg)	8466	230,071	10,371	256,182	1905	0.683	21,750	381,807
No. of destinations	2.7	4.3	2.9	4.5	0.1	0.215	4.2	6.1
No. of products (4-digit)	5.3	9.4	4.7	8.2	-0.6	0.000	5.1	8.7
No. of Prod x Dest	8.7	20.4	8.0	18.5	-0.7	0.047	10.9	23.4
Fraction of debt from banks with foreign debt above 10%	0.247	0.398	0.295	0.417	0.047	0.000	0.514	0.417
Banks' foreign debt weighted by firm's debt	0.037	0.036	0.041	0.036	0.005	0.000	0.047	0.027
Panel 2: Firm-product-destination	(N = 53,690)		(N = 47,810)				(N = 35,621)	
Value - FOB (1000 US\$)	169.3	3675.6	184.8	3875.5	15.4	0.516	245.0	4481.5
Volume (1000 kg)	401.7	22,112.8	446.4	23,432.3	44.6	0.756	593.2	27,139.0
Distance (km)	6521	7696	6518	8005	-4	0.943	6508	8,718
Paid in advance (FOB 1000 US\$)	42.2	1764.5	46.3	1869.6	4.2	0.716	61.5	2165.7
Transported by Air (FOB 1000 US\$)	37.2	2181.9	40.8	2311.6	3.6	0.797	52.9	2666.6
Shipment Value - FOB (1000 US\$)	32.8	470.2	36.0	497.1	3.2	0.294	47.5	574.9
Shipment Volume - (1000 kg)	77.5	2203.1	86.1	2334.4	8.5	0.550	113.9	2702.7
No. of Shipments per year	2.23	2.19	2.17	2.15	-0.06	0.000	2.39	2.34

TABLE 3
Transmission of credit shocks by banks

Dependent variable	$\Delta \ln C_{ib}$	
	(1)	(2)
FD_b	-1.881* (1.008)	
$D(FD_b > 10\%)$		-0.168*** (0.046)
Firm FE	Yes	Yes
Observations	10,334	10,334
R^2	0.630	0.630
R^2 adj	0.261	0.261
No. of banks	41	41
No. of firms	4974	4974

- Describes shipment-level features aggregated to the annual flow level: distance, whether paid in advance, transported by air, etc.
- These variables later appear as mechanisms and as potential sources of bias in aggregated regressions.

8.3 Table 3: transmission of credit shocks by banks

Model. First-differenced within-firm bank lending equation (4.8), with $f(FD_b)$ either linear or a high-exposure indicator.

Interpretation.

- The negative coefficient on FD_b indicates banks with more foreign funding reduce lending more after the reversal.
- The indicator specification implies that banks above the 10% threshold cut lending substantially relative to below-threshold banks.

TABLE 4
First stage with different instrumental variable functional forms

Dependent variable	$\Delta \ln C_i$		
	(1)	(2)	(3)
$\sum_b \omega_{ib} \times D(FD_b > \bar{FD})$	-0.991*** (0.092)		
$\sum_b \omega_{ib} \times FD_b$		-6.001*** (1.324)	
$\sum_b \omega_{ib} \times FD_b^2$			33.9*** (6.6)
$\sum_b \omega_{ib} \times FD_b^3$			-689.3*** (120.3)
			2863.7*** (510.9)
Product-destination FE	Yes	Yes	Yes
Observations	14,208	14,210	14,211
R^2	0.371	0.353	0.394
F -Statistic (1, 5995)	115.52	20.53	
F -Statistic (3, 5995)			13.78

- Because the regression includes firm fixed effects, the coefficient is interpreted as a *supply-side* contraction rather than firms' demand.

Connection. This table establishes the *relevance* of FD_b for credit supply, motivating F_i as an instrument.

8.4 Table 4: first stage with alternative instrument functional forms

Model. First stage (4.9) using different choices for F_i : binary indicator exposure, linear FD_b , and a polynomial in FD_b .

Interpretation.

- The instrument strongly predicts changes in firm credit: firms more exposed to high- FD_b banks experience larger credit declines.
- F -statistics show the binary instrument is particularly strong; linear exposure is weaker.
- Together with the LASSO exercise, this supports the baseline use of the threshold-based F_i in (4.10).

8.5 Table 5: intensive margin elasticity to credit shocks

Model. Second stage (4.11) estimated by OLS and IV, plus heterogeneity checks.

- **Column (1): OLS.** Regress $\Delta \ln X_{ipd}$ on $\Delta \ln C_i$ with product-destination FE. The estimate is small.
- **Column (2): IV.** Instrument $\Delta \ln C_i$ with F_i . The elasticity rises sharply (economically meaningful).
- **Columns (3)–(6): LATE representativeness.** Restrict to flows with high continuation probability and re-estimate; elasticities remain similar.
- **Column (7): continuation Probit.** Predict probability of continuation with pre-period export size and firm size/credit; importantly, the instrument does not directly shift continuation once credit is controlled for, supporting exclusion.

TABLE 5
Intensive margin elasticity to credit shocks

Dependent variable	$\Delta \ln X_{ipd}$						$Pr(X_{ipdPost} > 0 X_{ipdPre} > 0)$
	IV						Probit
	OLS (1)	Full sample (2)	$Pr > 0.42$ (3)	$Pr > 0.56$ (4)	$Pr > .68$ (5)	$Pr > .79$ (6)	(7)
$\Delta \ln C_i$	0.025** (0.010)	0.195*** (0.046)	0.162*** (0.052)	0.146** (0.072)	0.111 (0.105)	0.161 (0.134)	
$\ln X_{ipd}$							0.145*** (0.003)
$\ln X_i$							0.040*** (0.004)
$\ln C_i$							0.018*** (0.002)
$\sum_b \omega_{ib} \times D(FD_b > \overline{FD})$							-0.029 (0.022)
Product-Destination FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,208	14,208	10,708	8031	5354	2677	29,763
R^2	0.438						0.166

TABLE 6
Extensive margin elasticity to credit shocks

Dependent variable	$Pr(X_{ipdt} = 0 X_{ipdt-1} > 0)$ Exit		$Pr(X_{ipdt} > 0 X_{ipdt-1} = 0)$ Entry	
	OLS (1)	IV (2)	OLS (3)	IV (4)
$\ln C_{it}$	-0.001 (0.003)	-0.04 (0.011)	0.015 · 10 ⁻² (0.010 · 10 ⁻²)	-0.001 · 10 ⁻² (0.020 · 10 ⁻²)
Product-destination-time FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	62,386	62,386	4,161,4651	4,161,4651
R^2	0.591		0.520	

8.5.1 Why IV > OLS: a transparent attenuation-bias derivation

A convenient way to interpret the gap is through a “demand-as-measurement-error” decomposition. Suppose

$$\Delta \ln C_i = s_i + d_i,$$

where s_i is supply-driven credit variation and d_i is demand-driven variation (uncorrelated with s_i). If the true intensive-margin relationship is

$$\Delta \ln X_{ipd} = \eta^I s_i + u_{ipd},$$

then OLS on $\Delta \ln C_i$ yields

$$\hat{\eta}^{OLS} \approx \eta^I \cdot \frac{\text{Var}(s_i)}{\text{Var}(s_i + d_i)}.$$

Thus,

$$\frac{\eta^I}{\hat{\eta}^{OLS}} = \frac{\text{Var}(s_i) + \text{Var}(d_i)}{\text{Var}(s_i)} = 1 + \frac{\text{Var}(d_i)}{\text{Var}(s_i)}. \quad (12)$$

A large IV/OLS ratio implies demand factors explain a large share of total credit variation, so OLS is strongly attenuated.

TABLE 7
Effect of credit on export arrangements

Dependent variable	$\Delta \ln(\text{ShipFreq}_{ipd})$ (1)	$\Delta \ln(\text{ShipVol}_{ipd})$ (2)	$\Delta \ln(\text{ShipVal}_{ipd})$ (3)	$\Delta \ln(\text{FracAir}_{ipd})$ (4)	$\Delta \ln(\text{FracCash}_{ipd})$ (5)
$\Delta \ln C_i$	0.110*** (0.020)	0.085** (0.036)	0.106*** (0.037)	0.008 (0.007)	-0.024** (0.011)
Product-destination FE	Yes	Yes	Yes	Yes	Yes
Observations	14,208	14,208	14,209	14,208	14,208

8.6 Table 6: extensive margin elasticity (exit and entry)

Model. Linear probability specification (3.4) for exit and entry, estimated by OLS and IV.

Interpretation.

- The coefficients are small and not robustly distinguishable from zero in the paper's inference.
- Economically, this suggests that the credit shock does not primarily operate through entry/exit of export relationships (which are plausibly associated with sunk costs).
- This complements Table 5: credit affects *how much* firms export in ongoing markets, not *whether* they enter/exit markets.

8.7 Table 7: export arrangements and trade credit

Model. IV estimation of (4.11) with alternative dependent variables capturing shipment policies and payment terms.

Interpretation.

- When credit falls, firms reduce shipment frequency and shipment size (both in volume and value): credit matters for operating/working-capital decisions.
- No strong substitution toward air shipping is detected.
- The fraction paid in advance (a trade credit / payment term margin) moves in the expected direction but with economically small magnitude.

8.8 Table 8: identification and robustness tests

Purpose. Stress-test the identifying assumptions and show stability of the intensive-margin estimate.

Reading guide.

- Alternative export measures (value vs volume) and alternative product definitions (e.g., 6-digit HS) preserve the intensive-margin estimate.
- Restricting to more homogeneous goods yields similar elasticities, reducing concerns about within-product quality shifts.
- Adding observable firm characteristics addresses the concern that exposed firms differ systematically in ways not mediated by credit.

TABLE 8
Identification tests

Dep. variable	$\Delta \ln X_{ipd}$ Intensive (1)		$\Pr(X_{ipdt} = 0 X_{ipdt-1} > 0)$ Exit (2)	$\Pr(X_{ipdt} > 0 X_{ipdt-1} = 0)$ Entry (3)
Panel 1: X_{ipdt} is value (FOB) of exports				
$\Delta \ln C_i$	0.217*** (0.046)	$\ln C_{it}$	–	–
Panel 2: product classification at 6 HS digits				
$\Delta \ln C_i$	0.221*** –0.046	$\ln C_{it}$	–0.001 (0.009)	0.000 · 10 ^{–2} (0.000 · 10 ^{–2})
Panel 3: sample of homogeneous goods				
$\Delta \ln C_i$	0.183*** (0.048)	$\ln C_{it}$	–0.011 (0.012)	–0.001 · 10 ^{–2} (0.021 · 10 ^{–2})
Panel 4: controlling for observable firm characteristics				
$\Delta \ln C_i$	0.219*** (0.057)	$\ln C_{it}$	–0.016 (0.013)	–0.006 · 10 ^{–2} (0.022 · 10 ^{–2})
$\ln X_i$	–0.028** (0.013)	$\ln X_i \times Post$	–0.012*** (0.004)	–0.009* · 10 ^{–2} (0.005 · 10 ^{–2})
$\ln US\$Debt_i$	0.119* (0.065)	$\ln US\$Debt_i \times Post$	–0.030* (0.018)	0.024 · 10 ^{–2} (0.024 · 10 ^{–2})
$UnitPrice_{ipd}$	0.000 (0.000)	$UnitPrice_{ipd}$	0.000 (0.000)	
		$UnitPrice_{ipd} \times Post$	0.000 (0.000)	
$\ln \#products_i$	0.003 (0.022)	$\ln \#prod_i \times Post$	–0.015** (0.007)	–0.077*** · 10 ^{–2} (0.013 · 10 ^{–2})
$\ln \#destination_s$	0.034 (0.029)	$\ln \#dest_i \times Post$	–0.031*** (0.009)	–0.018 · 10 ^{–2} (0.022 · 10 ^{–2})
Panel 5: post-period of two years				
$\Delta \ln C_i$	0.189*** (0.038)	$\ln C_{it}$	–0.014 (0.010)	0.001*** · 10 ^{–2} (0.000 · 10 ^{–2})
Panel 6: measure of exposure as of Dec 2007				
$\Delta \ln C_i$	0.162*** –0.054	$\ln C_{it}$	–0.015 (0.013)	0.006 · 10 ^{–2} (0.022 · 10 ^{–2})
Panel 7: alternative IV functional form (Continuum 3rd-Degree Polynomial)				
$\Delta \ln C_i$	0.217*** (0.064)	$\ln C_{it}$	–0.028* (0.016)	0.007 · 10 ^{–2} (0.027 · 10 ^{–2})
Panel 8: placebo test				
$\Delta \ln C_i$	–0.125 (0.173)	$\ln C_{it}$	–0.077 (0.057)	0.001 · 10 ^{–2} (0.001 · 10 ^{–2})

- A placebo timing test (moving the “shock” to an earlier period) yields no effect, supporting the event-time interpretation.

8.9 Table 9: estimation bias from aggregation and omitted shocks

Purpose. Demonstrate why product–destination controls are crucial.

Design. Interact credit with proxies such as distance, air shipping intensity, and cash-in-advance share, estimated with and without product–destination fixed effects.

Interpretation.

- Without product–destination controls, interactions can appear significant, spuriously suggesting heterogeneity.

TABLE 9
Estimation bias

Dependent variable	$\Delta \ln X_{ipd}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \ln C_i$	0.082 (0.059)	0.195*** (0.046)	0.098** (0.041)	0.177*** (0.053)	0.111*** (0.037)	0.195*** (0.046)	0.093** (0.039)	0.182*** (0.048)
$\Delta \ln C_i \times Dist_d$			–0.028 (0.094)	–0.062 (0.055)				
$\Delta \ln C_i \times Air_{ipd}$					0.065* (0.038)	0.068 (0.047)		
Air_{ipd}					–0.084*** (0.022)	–0.132*** (0.033)		
$\Delta \ln C_i \times Cash_{ipd}$							–0.077* (0.039)	–0.082 (0.056)
$Cash_{ipd}$							0.042 (0.028)	0.031 (0.041)
Prod-dest FE	No	Yes	No	Yes	No	Yes	No	Yes
Observations	14,208	14,208	14,146	14,146	14,208	14,208	14,208	14,208

TABLE 10
Elasticity by product characteristic

Dependent variable	$\Delta \ln X_{ipdt}$ Intensive margin (1)	$\Pr(X_{ipdt} = 0 X_{ipdt-1} > 0)$ Exit (2)	$\Pr(X_{ipdt} > 0 X_{ipdt-1} = 0)$ Entry (3)
$\Delta \ln C_i$	0.164*** (0.049)		
$\Delta \ln C_i \times HighFinDep_p$	-0.132 (0.084)		
$\ln C_{it}$		0.008 (0.014)	$0.006 \cdot 10^{-2}$ ($0.021 \cdot 10^{-2}$)
$\ln C_{it} \times HighFinDep_p$		0.065 (0.055)	$0.001 \cdot 10^{-2}$ ($0.004 \cdot 10^{-2}$)
Product-destination FE	Yes	No	No
Product-destination-time FE	No	Yes	Yes
Firm FE	No	Yes	Yes
Observations	14,208	56,215	4,119,771

- Once product–destination shocks are controlled for, the interactions vanish.
- Lesson: cross-sectional comparisons across products/destinations (or sector-level exports) can severely confound credit effects with non-credit shocks.

8.10 Table 10: elasticity by product external-finance dependence

Purpose. Test whether credit effects are stronger in sectors more dependent on external finance (Rajan–Zingales style).

Interpretation.

- The interaction between credit and “high external-finance dependence” is not large or precisely estimated in the paper’s setting.
- This result is consistent with the paper’s central message: once one controls for product–destination shocks at high resolution, sectoral “finance dependence” patterns need not map cleanly into differential export responses.

9 Summary

- Tables 1–4 and Figures 1–2 establish the *credit supply shock*: foreign funding dependence predicts differential bank lending contraction, and firm exposure to such banks predicts firm credit contraction.
- Table 5 and Figure 3 establish the *intensive margin channel*: continuing export flows scale down with credit; the estimate is robust and not driven by composition.
- Table 6 establishes that the *extensive margin* is weak: entry/exit is not the main adjustment channel.
- Table 7 pins down *micro adjustment*: fewer/smaller shipments, modest changes in payment terms.
- Tables 8–10 show robustness and warn against *aggregation bias*: the product–destination dimension is essential for valid inference.